25900656 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course (VAC) - MG3VACSTA200 - APPLIED STATISTICAL ANALYSIS : ETHICAL DATA COLLECTION, INTERPRETATION AND DECISION MAKING IN SOCIETY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer type questions

Answer any **7** questions

Each question carries **2** marks

1. What does coefficient of correlation r = 0 imply? [K] / [CO1]
2. What is Simple Linear Regression? [K] / [CO1]
3. Define mode. [U] / [CO1]
4. What are ogives? [K] / [CO1]
5. Define chi square test. [U] / [CO2]
6. State the test statistic for testing the difference of two population means in large samples. [U] / [CO2]
7. Define statistical inference. [K] / [CO2]
8. Which value of correlation indicates a perfect positive relationship? [U] / [CO1]
9. Which statistical measure gives the average value of a dataset? [U] / [CO1]
10. What is the key ethical requirement you must secure from a participant before an interview, often documented on a form? [An] / [CO3]

[2x7 = 14]

Part B

Short Essay Type Questions

Answer any **4** questions

Each question carries **6** marks

11. How is "association" between two attributes defined, and differentiate between positive and negative association using the delta (δ) notation? [K] / [CO1]

12. Discuss the four measures of scaling of data. [K] / [CO1]

13. For a dichotomous classification with 'n' attributes, provide the formula to determine the number of class frequencies of order 'r'. [K] / [CO1]

14. In 120 throws of a six faced die the even numbers occur 55 times. Is the die unbiased (Level of significance =0.05)? [A] / [CO2]

15. A sample of 36 students with mean mark 52 and standard deviation 6 is taken. Test, at 5% significance level, whether the college's claim that the average mark of students is 50 is true. [A] / [CO2]

16. Explain the role of the p-value in interpreting the result of a Z test. [U] / [CO3]

[6x4 = 24]

Part C

Essay type questions

Answer any **1** questions

Each question carries **12** marks

17. Find combined Standard deviation(SD) based on the following data. [A] / [CO1]

Group	No. of Items	Mean	SD
A	10	2.5	2
B	5	3	3
C	5	2	5

18. Two random samples are taken from normal populations resulting in the following statistic. First sample Size :16,. Sample mean: 34 and standard deviation: 2.5. Second sample size 25, sample mean 45 and standard deviation 2.5 respectively . Test whether the samples can be regarded as coming from the same normal population at 5% level of significance. [A] / [CO2]

[12x1 = 12]